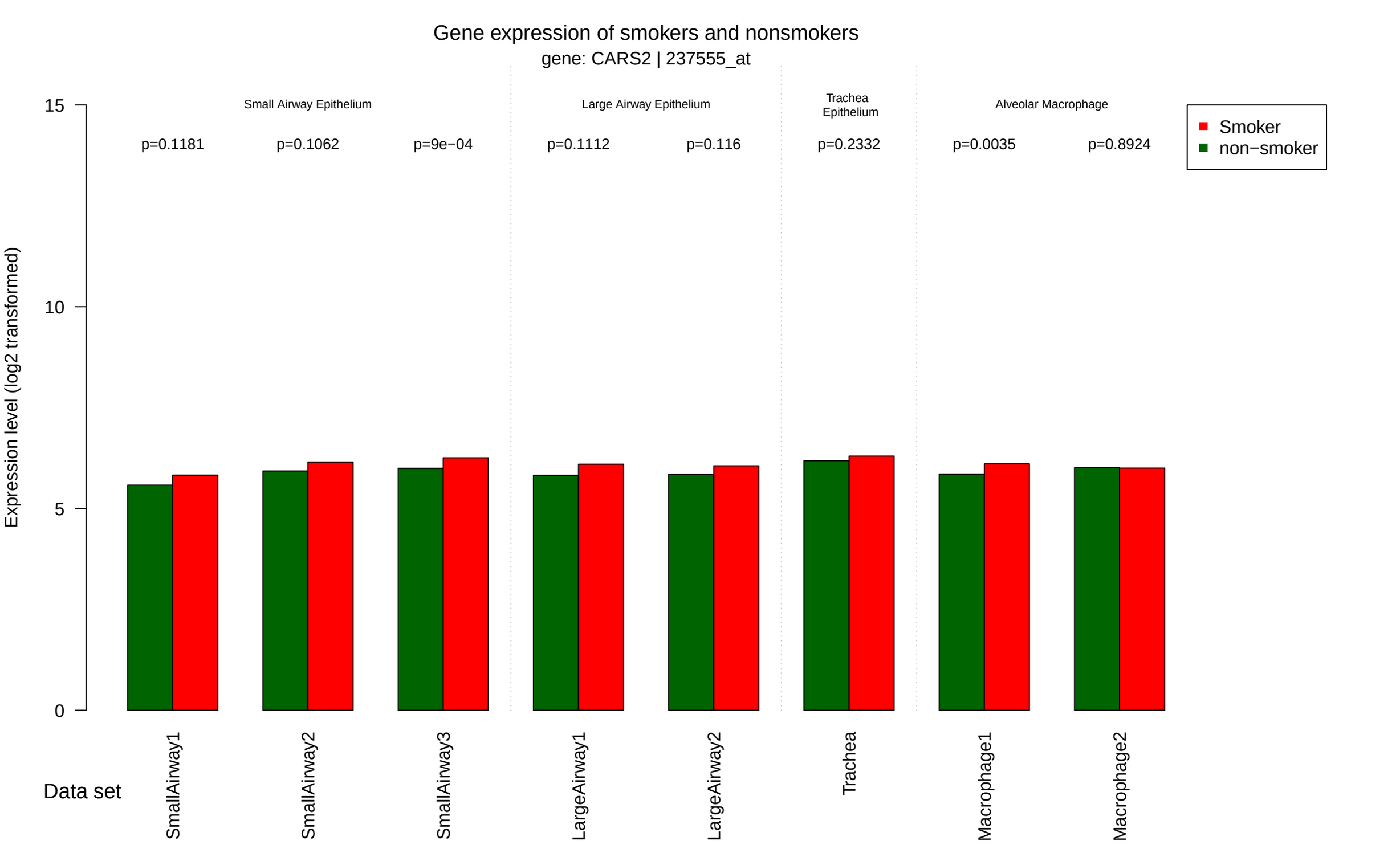
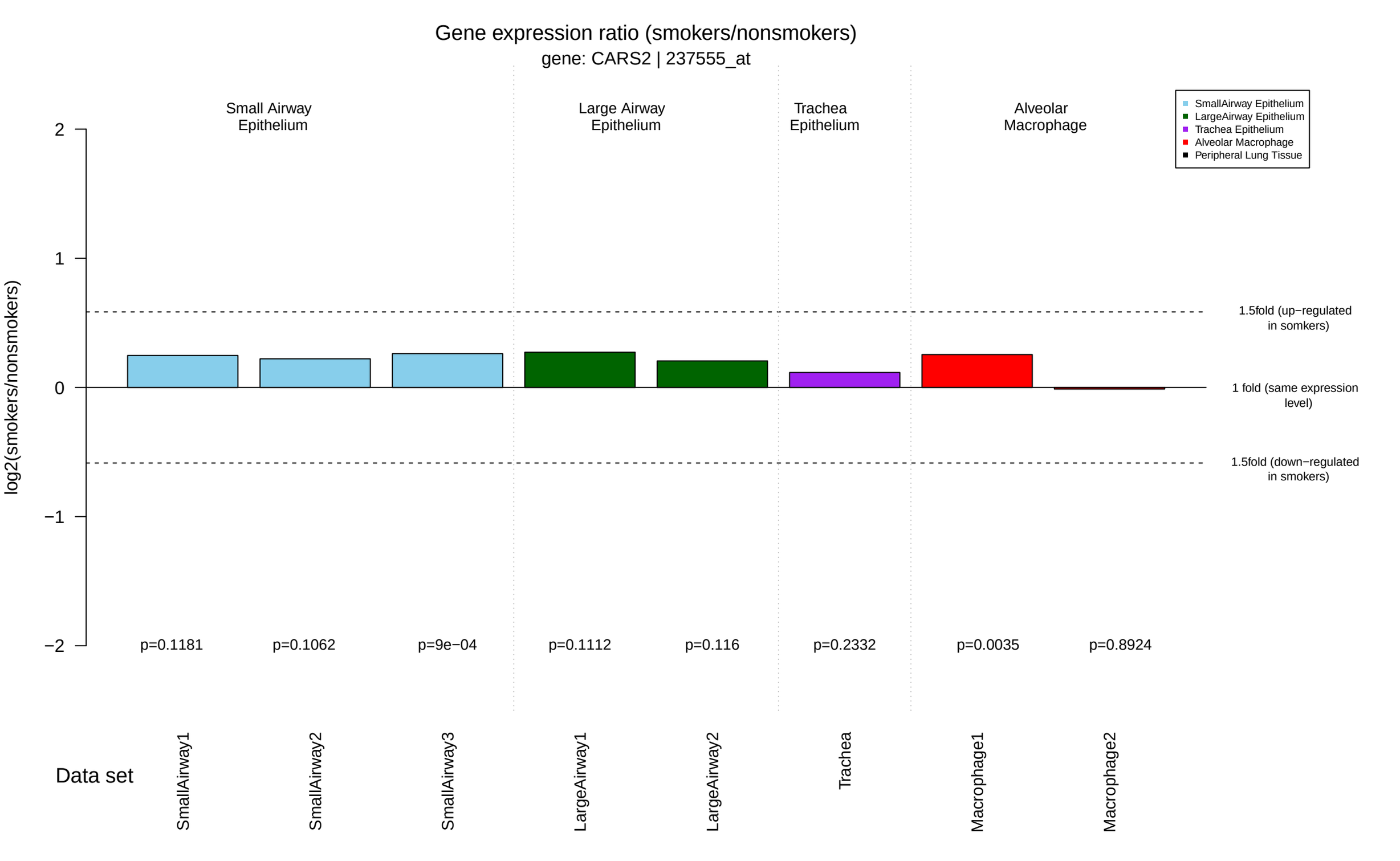
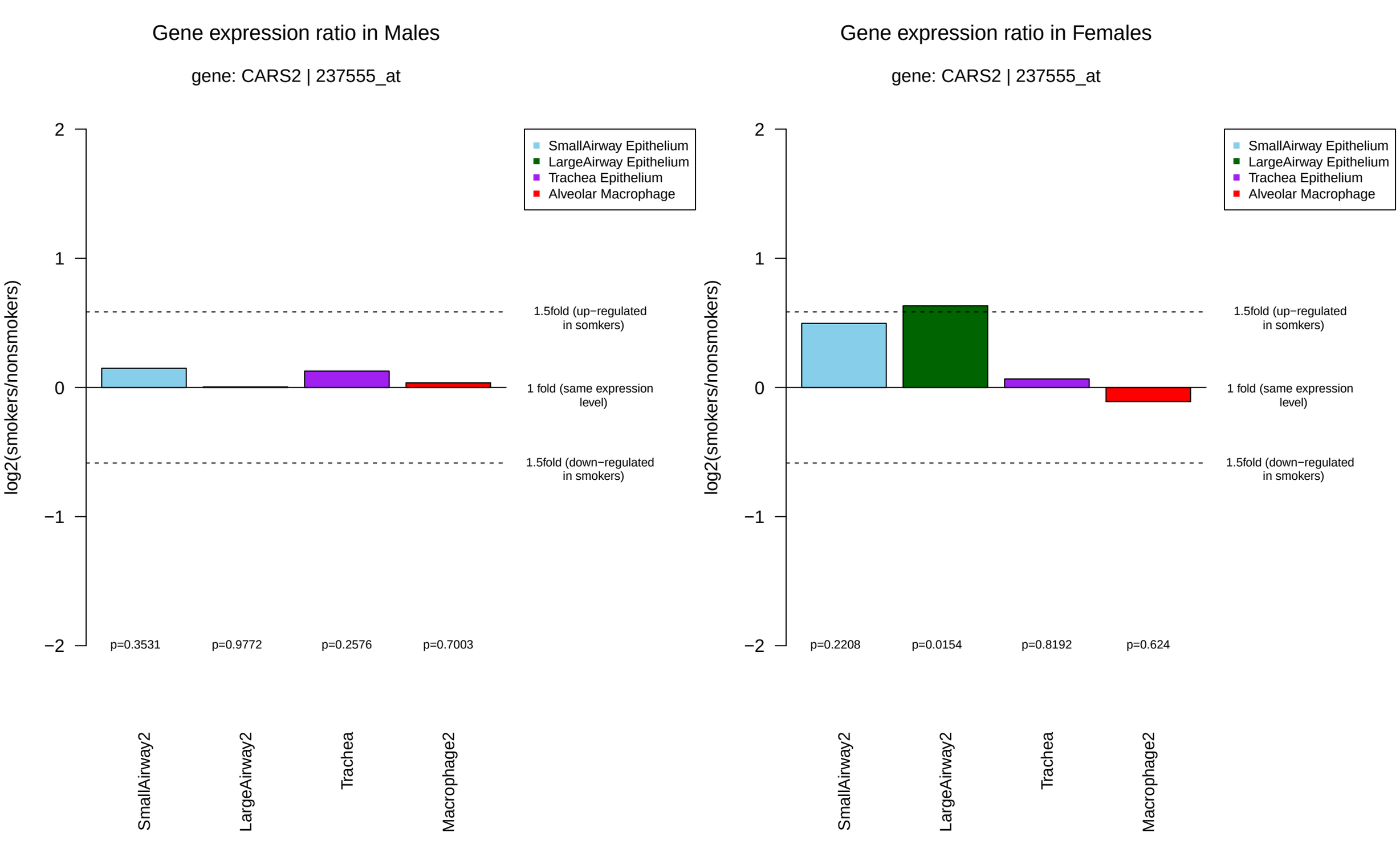
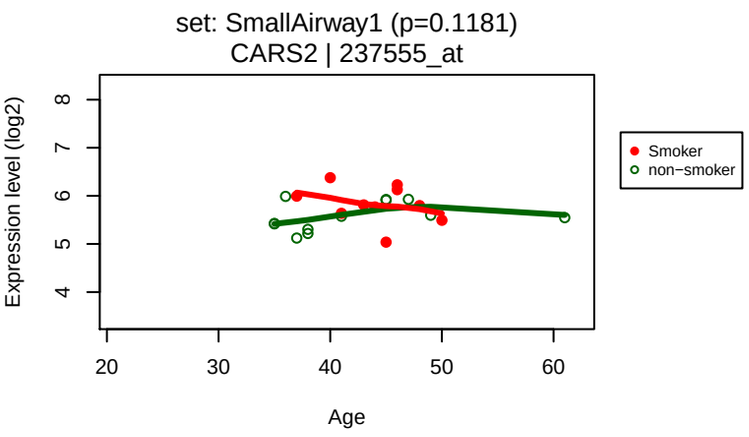
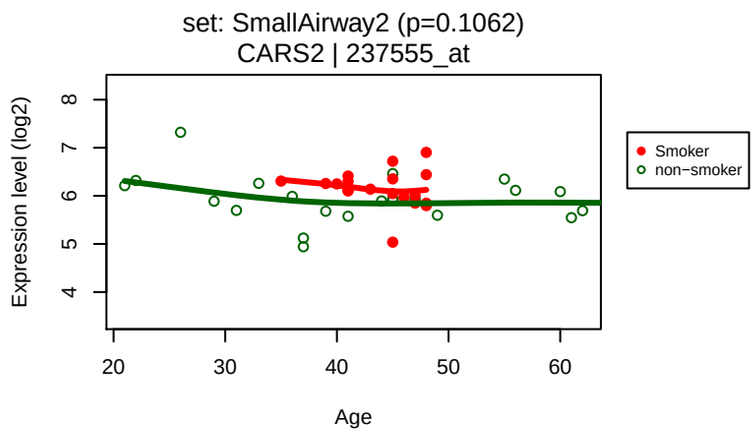
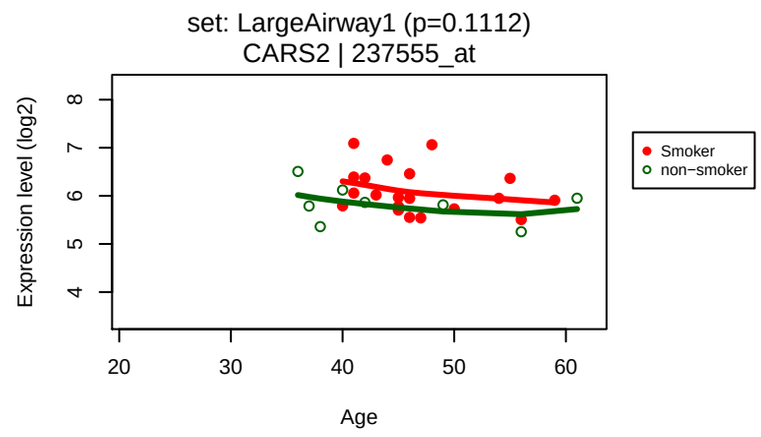

Gene expression of smokers and nonsmokers
gene: CARS2 | 237555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1181
p=0.1062
p=9e−04
p=0.1112
p=0.116
p=0.2332
p=0.0035
p=0.8924
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

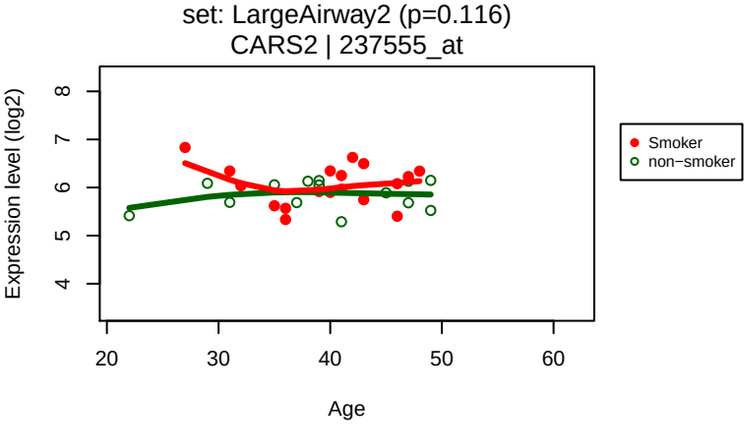

Gene expression ratio (smokers/nonsmokers)
gene: CARS2 | 237555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1181
p=0.1062
p=9e−04
p=0.1112
p=0.116
p=0.2332
p=0.0035
p=0.8924
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

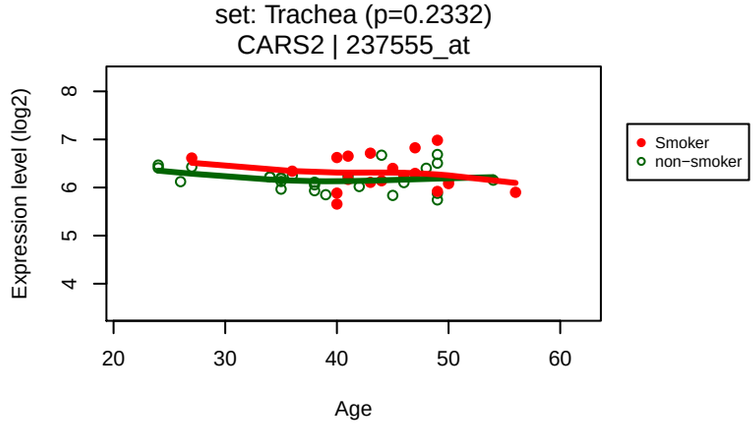

Gene expression ratio in Males
gene: CARS2 | 237555_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3531
p=0.9772
p=0.2576
p=0.7003
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CARS2 | 237555_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2208
p=0.0154
p=0.8192
p=0.624
SmallAirway2
LargeAirway2
Trachea
Macrophage2

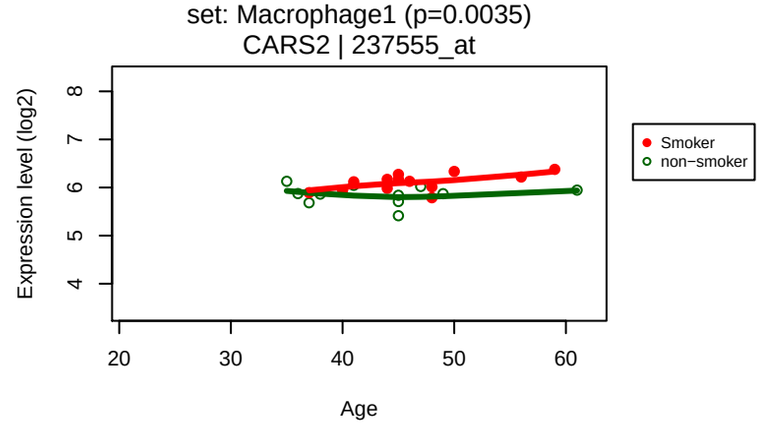

set: SmallAirway1 (p=0.1181)
CARS2 | 237555_at
Expression level (log2)
Age
Smoker
non−smoker

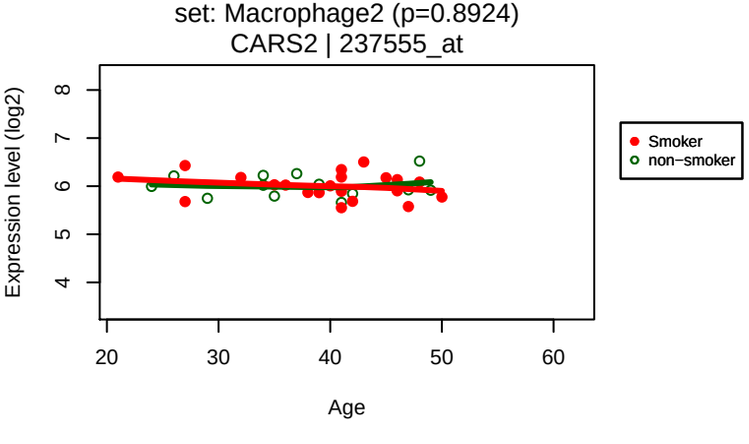

set: SmallAirway2 (p=0.1062)
CARS2 | 237555_at
Expression level (log2)
Age
Smoker
non−smoker

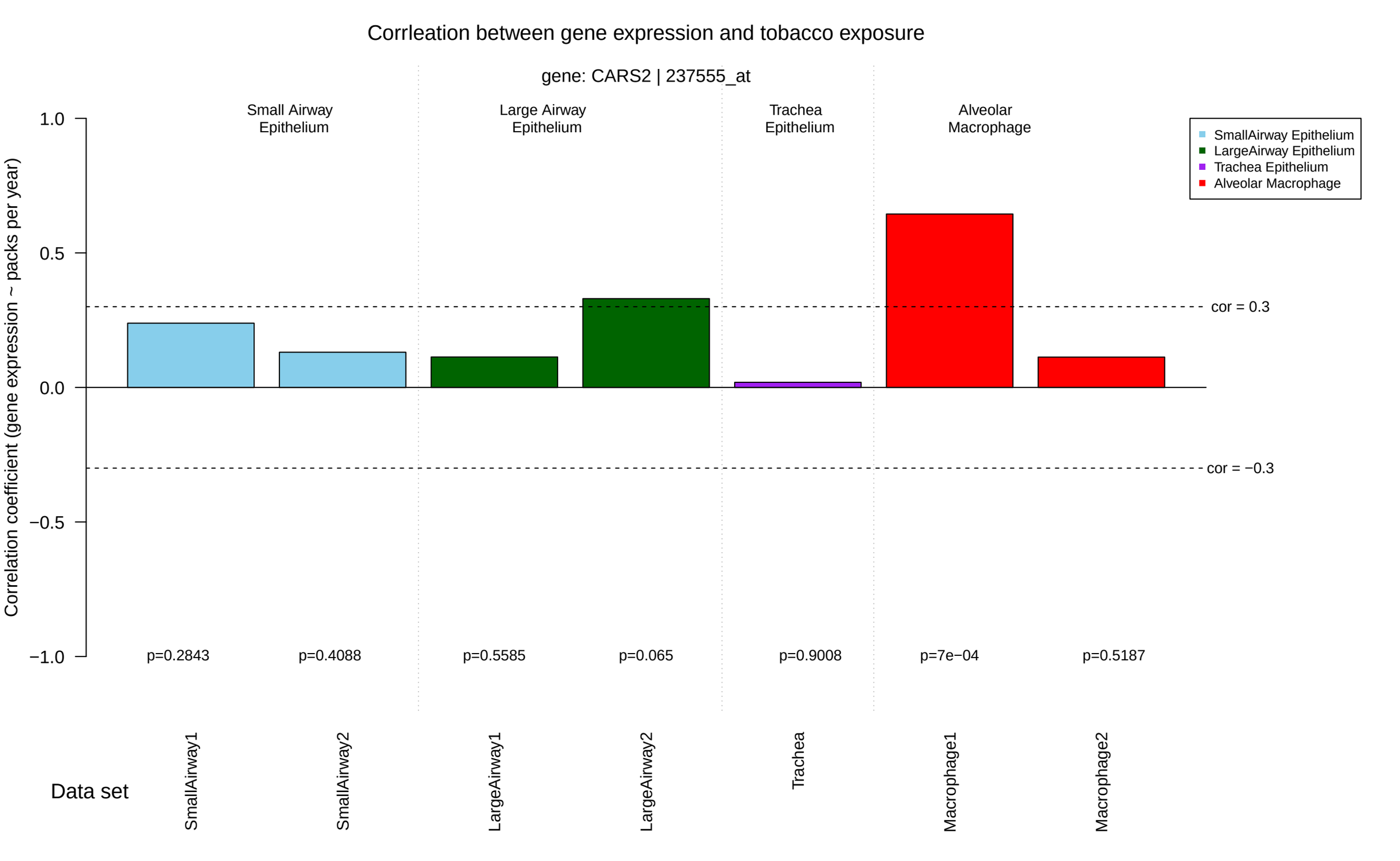

set: LargeAirway1 (p=0.1112)
CARS2 | 237555_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.116)
CARS2 | 237555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2332)
CARS2 | 237555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0035)
CARS2 | 237555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8924)
CARS2 | 237555_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CARS2 | 237555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2843
p=0.4088
p=0.5585
p=0.065
p=0.9008
p=7e−04
p=0.5187
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2